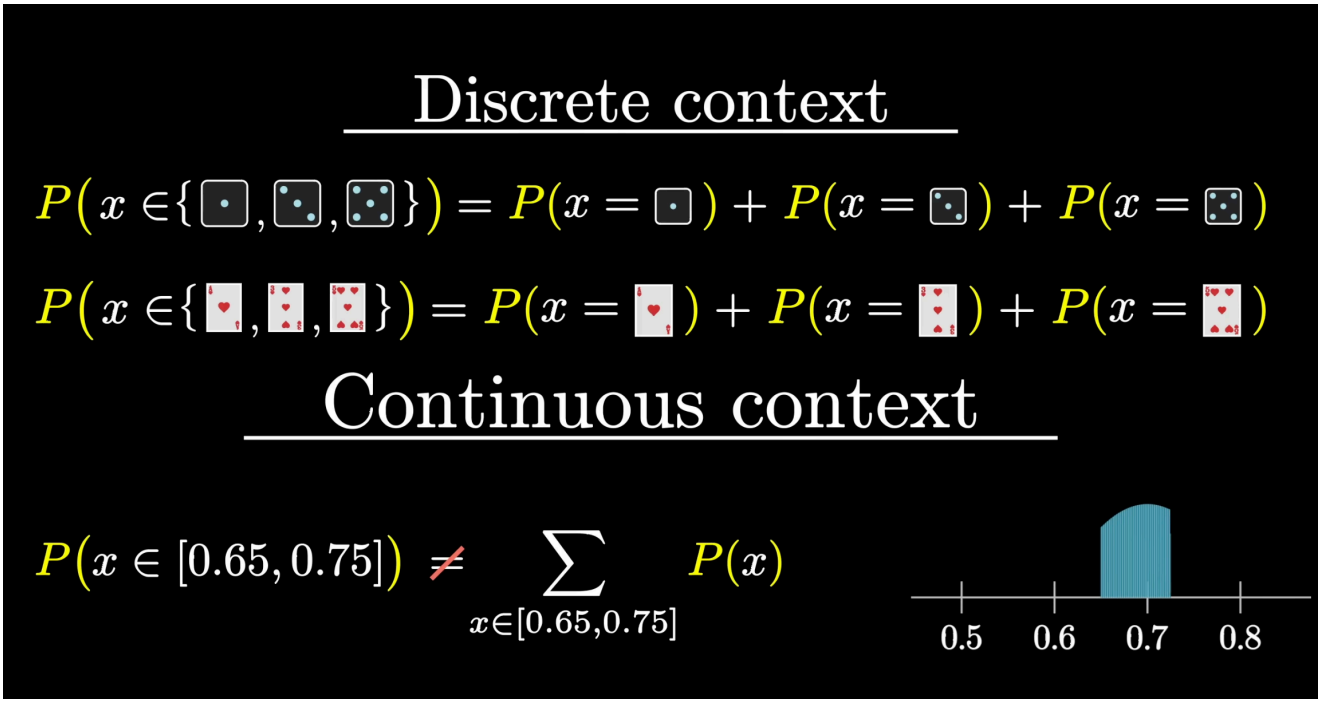
PDF (Probability Density Function)



A **PDF** describes how **likely a continuous random variable** is to take values **within a range**.

Key idea:

- It applies to **continuous** data (not discrete)
- It does **NOT** give probability at a single point
- Probability comes from the **area under the curve**

Key Points about PDF Formula

- *If we differentiate CDF, we get the PDF of the random variable.*

$$f(y) = \frac{d}{dy}[F(y)]$$

- *If we integrate PDF, we get the CDF of the random variable.*

$$F(y) = \int_{-\infty}^{y} f(t)dt$$

Formal definition

For a continuous random variable (X), a function (f(x)) is a PDF if

1. $f(x) \geq 0$ for all x
2. Total area under the curve is 1

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

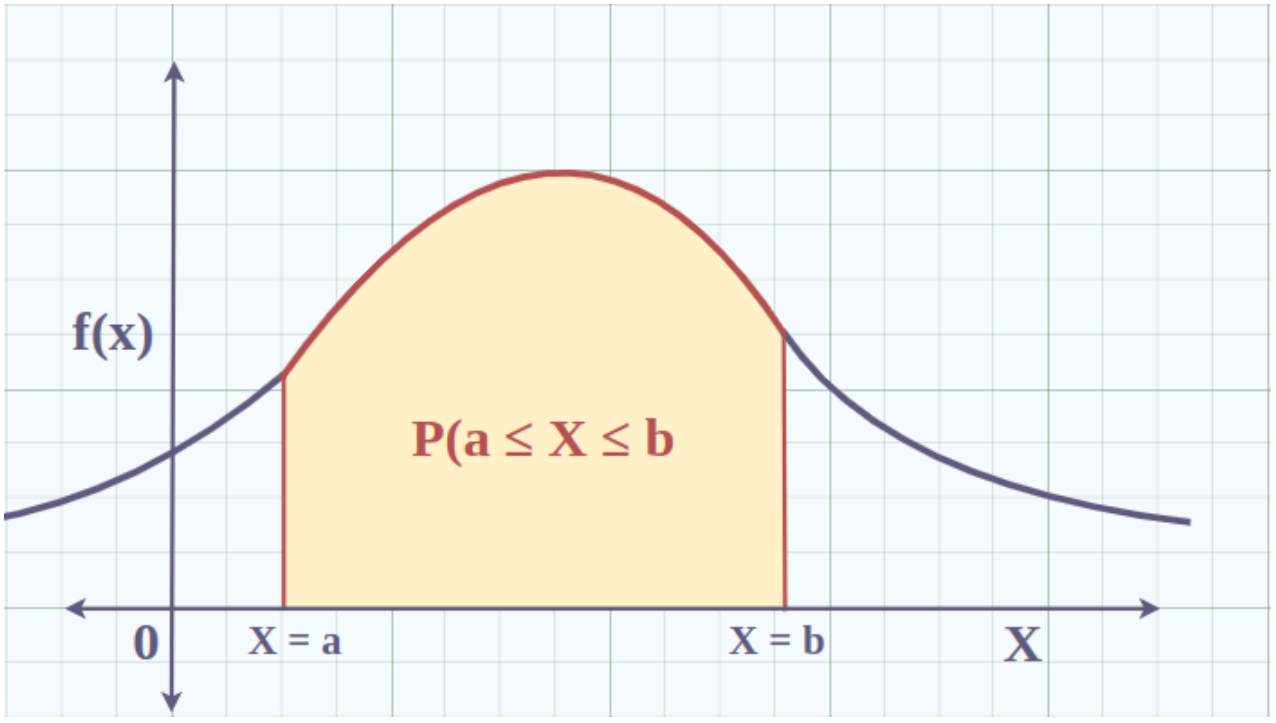
3. Probability over an interval:

$$P(a \leq X \leq b) = \int_{a}^{b} f(x)\,dx$$

Important

(P(X = c) = 0) for any continuous random variable
Why?

- A single point has **zero width**
- Probability only exists over **intervals**



Simple example: Uniform distribution

Suppose:

- $X \sim \text{Uniform}(0, 10)$

PDF:

$$f(x) = \begin{cases} \frac{1}{10}, & 0 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

Probability example

Find $P(2 \leq X \leq 5)$:

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Classic example: Normal distribution

If
$X \sim N(\mu, \sigma^2)$

PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Used for:

- Heights
- Test scores
- Measurement errors

Smooth, bell-shaped curve 🔔

PDF vs PMF

FEATURE	PDF	PMF
Data type	Continuous	Discrete
Used for	Normal, Exponential	Binomial, Poisson
Single value probability	Always 0	Can be > 0
Probability comes from	Area	Sum

Intuition (plain English)

- PDF height = **density**, not probability
- Probability = **area under the curve**
- Taller curve ≠ more likely at a point
- Wider interval = more probability